

Divergence Detector

Indicators & Analysis / Levels & Patterns | advanced | 6 min read | DeepCharts Help Center | July 15, 2026

A breakout that nobody participates in is a breakout worth doubting. The Divergence Detector watches for exactly that: it highlights volume or delta divergence with labels directly on the chart, flagging moments when a new relative high or low is made without a corresponding increase in volume or delta.

Because thin-participation extremes often precede false breaks and reversals, the indicator acts as an automated skeptic — it marks the highs and lows where the effort behind the move did not match the result.

What it is

The Divergence Detector compares price extremes against the volume and delta that produced them. When price prints a new relative high or low (identified through a built-in Zig Zag swing engine) but volume or delta fails to expand with it, the indicator prints a label at that extreme:

- **VD** — volume divergence only.
- **DV** — delta divergence only.
- **VDD** — both volume and delta divergence, with the most significant divergence prioritized.

Delta here is the difference between aggressive buying and aggressive selling — if that concept is new, read [Orderflow 101: Bid, Ask, Aggressor and Delta](#) first.

When to use it

- To fade suspect breakouts: a new high on shrinking volume or weak delta is a classic false-break setup.
- To time exits: divergence at an extreme in your favor is a reason to tighten stops or scale out, even if you don't reverse.
- To confirm swing failures: combined with a structure read from Swing Point or Zig Zag, a labeled divergence adds evidence that a swing will not hold.
- To monitor participation in real time — the Tick calculation mode evaluates divergence the moment a previous extreme breaks, not only at bar close.

Quick start

1. Open a chart and click the bar-chart icon in the top-left corner to open the **Indicators** panel.
2. Click **Indicators** to open the full **Indicator List**.
3. Search for "Divergence Detector" and click **+** to add it.
4. Click the gear icon next to the indicator to open its settings. Set **Calculation Mode (1)** to **Volume and delta** so you see the full picture (VDD labels), and start with **Calculation Mode (2) = Close** so signals only print on completed candles.
5. Tune the Zig Zag settings so swings match the structure you actually trade — if labels appear on every minor wiggle, the reversal parameter is too small.

[SCREENSHOT PLACEHOLDER] A futures chart with the Divergence Detector active — a new swing high tagged with a VDD label above the candle, showing price making a higher high while participation faded

`divergence-detector-on-chart.png`

How to read it

- **A label at a new high** means price exceeded the prior relative high without a matching increase in volume (VD), delta (DV), or both (VDD). The move lacked participation — a warning that the breakout may fail.

- **A label at a new low** is the mirror: downside extension without expanding selling effort.
- **VDD outranks a single-source label.** When both volume and delta diverge at the same extreme, the evidence is stronger; the label prioritizes the most significant divergence.
- **Timing mode changes meaning.** In **Tick** mode a label can appear intrabar the moment the prior extreme breaks — earliest signal, most noise. **Close** mode waits for the candle to finish — latest, cleanest. **Sec. to close** is the compromise, evaluating a set number of seconds before the close.
- **Divergence is a warning, not an order.** Markets can grind to new extremes on weak participation for longer than a fade position survives; require a structural trigger before acting.

Settings reference

General

Setting	What it does
Calculation Mode (1)	Chooses which divergences display: Volume (VD labels only), Delta (DV labels only) or Volume and delta (VDD label, with the most significant divergence prioritized).
Calculation Mode (2)	Sets when divergence is evaluated: Tick (at the break of the previous high/low tick), Sec. to close (at a specified number of seconds before candle close) or Close (at candle close).
Maximum Seconds To Close	The timing parameter used by the Sec. to close mode.

Zig Zag Settings

The swing highs and lows the detector compares against are found by an internal Zig Zag.

Setting	What it does
Zig Zag Mode	How reversals (swing points) are identified: Absolute Reversal uses a fixed price reversal value, requiring sufficient price movement in the opposite direction; Tick Reversal is tick-based, allowing a minimum tick movement to generate a reversal; Highest Lowest is based on the tick difference between the swing maximum and minimum.
Absolute reversal parameter	The reversal value used in Absolute Reversal mode.
Tick Reversal / Highest Lowest parameter	The tick value used by the Tick Reversal and Highest Lowest modes.

Plot Settings

Setting	What it does
Text size	Size of the divergence labels.
Distance ticks	Distance, in ticks, between the label and the candle extreme it marks.

Alerts and Subgraphs

Setting	What it does
Alert Sound	Enables a sound notification when a divergence is detected.
Subgraphs	Customizes the label background and text colors.

[SCREENSHOT PLACEHOLDER] The Divergence Detector settings dialog showing Calculation Mode (1) and Calculation Mode (2) dropdowns, Maximum Seconds To Close, and the Zig Zag Settings group with the three reversal modes `divergence-detector-settings.png`

Tips and common mistakes

- **Pick the calculation timing that matches your style.** Scalpers watching for immediate failure want **Tick**; if you review completed candles, **Close** avoids labels that would have disappeared by the bar's end.
- **The Zig Zag defines everything.** If swings are defined too tightly, every micro-extreme gets tested for divergence and label count explodes; too loosely, and real reversals pass unlabeled. Tune the reversal parameter per instrument and timeframe.
- **Volume and delta divergence are not interchangeable.** Volume divergence says total activity faded; delta divergence says the aggressor imbalance faded. A high on strong volume but weak delta tells a different story than one on weak volume — running **Volume and delta** mode preserves that distinction.
- **Expect divergences to fail in strong trends.** Persistent trends routinely print weak-participation extremes and keep going; use divergence as a filter against chasing, not as a standalone reversal system.

Related articles

- Orderflow 101: Bid, Ask, Aggressor and Delta
- Zig Zag
- Delta Bar
- Volume
- Different Types of Input Data for Indicators
- How to Change the Layout of Indicators